

# DAX FUNCTIONS

## ✓ DAX (Data Analysis Expressions)

- **DAX** is a **formula language** used in Power BI (also in Power Pivot and SSAS).
- It's designed to **create custom calculations** on data after it's loaded into the data model.
- You use DAX to define **measures**, **calculated columns**, and **calculated tables**.

### Where is DAX Used?

- **Measures:** Calculations done on-the-fly (e.g., Total Sales)
- **Calculated Columns:** New columns added to a table using logic (e.g., Profit Margin)
- **Calculated Tables:** Entire tables generated based on DAX expressions

```
1 q_Growth_Subtitle =
2 VAR GrowthVal = [q_Growth]
3 RETURN
4 IF(
5     GrowthVal >= 0,
6     "▲ " & FORMAT(GrowthVal, "0.0") & " %",
7     "▼ " & FORMAT(ABS(GrowthVal), "0.0") & " %"
8 )
9
1 q_QuantitySP =
2 CALCULATE(
3     [q_totalQuantity],
4     FILTER(
5         'Sales Data',
6         'Sales Data'[OrderDate] IN VALUES('Calendar Lookup'[Date]
7     ))
```

X ✓ 1 Profit = 'Sales Data'[OrderQuantity]\*('Sales Data'[SP]-'Sales Data'[CP])

```
1 Sales_LP =
2 CALCULATE(
3     [s_Total],
4     FILTER(
5         'Sales Data',
6         'Sales Data'[OrderDate] IN
7         SELECTCOLUMNS(
8             ADDCOLUMNS(
9                 VALUES('Calendar Lookup'[Date]),
10                "PREYEARDATE",DATEADD('Calendar Lookup'[Date],-1,YEAR)
11            ),
12            "DATE",[PREYEARDATE]
13        )
14     )
15 )
```

- **Concept**
- **Filter Context**
- **Row Context**
- **CALCULATE()**
- **RELATED()**
- **Description**
- The set of filters applied to the data (e.g., a slicer on the report)
- The context of a single row in a table (used in calculated columns)
- Changes the filter context to perform custom calculations
- Fetches values from related tables

| Purpose           | DAX Formula                                            | Meaning                               |
|-------------------|--------------------------------------------------------|---------------------------------------|
| Total Revenue     | SUM(Sales[Revenue])                                    | Adds up all revenue                   |
| Revenue from Asia | CALCULATE(SUM(Sales[Revenue]), Sales[Region] = "Asia") | Sums revenue only for Asia            |
| Profit            | Sales[Revenue] - Sales[Cost]                           | Subtracts cost from revenue           |
| Average Sales     | AVERAGE(Sales[Amount])                                 | Calculates average sale               |
| Total Orders      | COUNTROWS(Orders)                                      | Counts number of rows in Orders table |